

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 11-2023: A 67-Year-Old Man with Mantle-Cell Lymphoma and Hypoxemia

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PRESENTATION OF CASE

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Dr. Samuel A. Hofacker (Medicine): A 67-year-old man was admitted to this hospital with hypoxemia.

Six years before the current presentation, the patient was evaluated because of melena and anemia. Esophagogastroduodenoscopy revealed duodenal ulcers; biopsy of one of the ulcers revealed changes consistent with moderately differentiated adenocarcinoma. Computed tomography (CT) of the abdomen showed a locally invasive mass in the head of the pancreas. A diagnosis of pancreatic adenocarcinoma with metastasis to the duodenum was made, and treatment included chemotherapy, chemoradiation, and pancreaticoduodenectomy. Two months after the surgery, surveillance CT of the chest showed segmental pulmonary embolism in the right lower lobe. Treatment with low-molecular-weight heparin was recommended, but the patient declined.

During the next 5 years, surveillance CT of the chest, abdomen, and pelvis was performed every 6 months. Six months before the current presentation, CT of the abdomen showed a new mass in the sigmoid colon with associated nodularity and lymphadenopathy. Colonoscopy revealed two fungating, nonobstructing, medium-sized masses, one at the ileocecal valve and the other in the sigmoid colon. Biopsy was performed, and a diagnosis of mantle-cell lymphoma was established. The patient was evaluated in the oncology clinic of this hospital, and plans were made for treatment with chemotherapy. Five months before the current presentation, pain in the right leg developed after a long car trip. Ultrasonography of the right leg revealed distal deep-vein thrombosis, and treatment with apixaban was initiated.

Nine weeks before the current presentation, cycle 1 of rituximab, dexamethasone, high-dose cytarabine, and cisplatin (R-DHAP) chemotherapy was administered, along with filgrastim. In addition, the administration of acyclovir and trimethoprim-sulfamethoxazole was started for prophylaxis; however, rash developed on the trunk after 1 week, and trimethoprim-sulfamethoxazole was replaced with atovaquone for prophylaxis against *Pneumocystis jirovecii* pneumonia.

Six weeks before the current presentation, cycle 2 of R-DHAP chemotherapy was

administered. Two days later, melena, fatigue, and dyspnea on exertion developed. Laboratory testing revealed anemia (hematocrit, 26.8%; normal range, 41.0 to 53.0). The patient received 2 units of packed red cells, and fatigue and dyspnea resolved. Treatment with omeprazole was initiated. Four weeks before the current presentation, the administration of atovaquone was stopped because of the cost of the medication, and dapsone was started for prophylaxis against *P. jirovecii* pneumonia.

Three weeks before the current presentation, cycle 3 of R-DHAP chemotherapy was administered. Ultrasonography of the right leg revealed resolution of the distal deep-vein thrombosis, and treatment with apixaban was stopped. Two weeks before the current presentation, fatigue and dyspnea on exertion recurred, but melena did not. Laboratory testing again revealed anemia (hematocrit, 28.7%), and the patient received 1 unit of packed red cells; fatigue resolved, but dyspnea did not.

On the day of presentation, the patient was evaluated before the administration of cycle 4 of R-DHAP chemotherapy. He reported ongoing dyspnea on exertion but no cough, fever, chills, chest pain, or orthopnea. He had been having loose stools since the pancreaticoduodenectomy. Other medical history included hypertension and hereditary spherocytosis, which had been treated with splenectomy. Medications included acyclovir, dapsone, lisinopril, and omeprazole. Trimethoprim-sulfamethoxazole had caused a rash; there were no other known drug allergies. The patient drank alcohol rarely; he had smoked tobacco for 2 years but had quit 20 years before this presentation. He was divorced, lived alone in a suburb of Boston, and worked as a lawyer. His mother had had breast cancer.

On examination, the temporal temperature was 36.6°C, the blood pressure 134/82 mm Hg, the pulse 90 beats per minute, and the respiratory rate 22 breaths per minute. The oxygen saturation was 92% while the patient was at rest and breathing ambient air, 88 to 90% while he was walking and breathing ambient air, and 93% while he was at rest and receiving supplemental oxygen through a nasal cannula at a rate of 6 liters per minute. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 25.4. The patient had

Table 1. Laboratory Data.

Variable	Reference Range, Adults*	On Presentation
Hematocrit (%)	41.0–53.0	25.2
Hemoglobin (g/dl)	13.5–17.5	7.9
White-cell count (per μ l)	4500–11,000	21,560
Differential count (per μ l)		
Neutrophils	1800–7700	14,380
Lymphocytes	1000–4800	750
Monocytes	200–1200	5280
Eosinophils	0–900	0
Basophils	0–300	190
Metamyelocytes	0	900
Platelet count (per μ l)	150,000–400,000	596,000
Mean corpuscular volume (fl)	80.0–100.0	107.7
Lactate dehydrogenase (U/liter)	110–210	382

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

tachypnea but no increased work of breathing. The lung sounds were normal, and the jugular venous pulse was not elevated. The legs had no edema.

The blood levels of glucose and electrolytes were normal, as were the results of coagulation, kidney-function, and liver-function tests. Other laboratory test results are shown in Table 1. Testing of a nasopharyngeal swab for severe acute respiratory syndrome coronavirus 2 RNA was negative, as was a respiratory viral panel. An electrocardiogram showed normal sinus rhythm.

Dr. Jo-Anne O. Shepard: CT pulmonary angiography of the chest (Fig. 1), performed after the administration of intravenous contrast material, showed filling defects in the right upper lobe and both lower lobes, findings consistent with pulmonary emboli. Subtle tree-in-bud and ground-glass opacities were present in the upper lobes. Scattered pulmonary nodules were present bilaterally; they were stable in size as compared with nodules observed on previous imaging studies. The main pulmonary artery measured 2.6 cm in diameter proximal to bifurcation.

Dr. Hofacker: Cycle 4 of R-DHAP chemotherapy was delayed. Treatment with intravenous hepa-

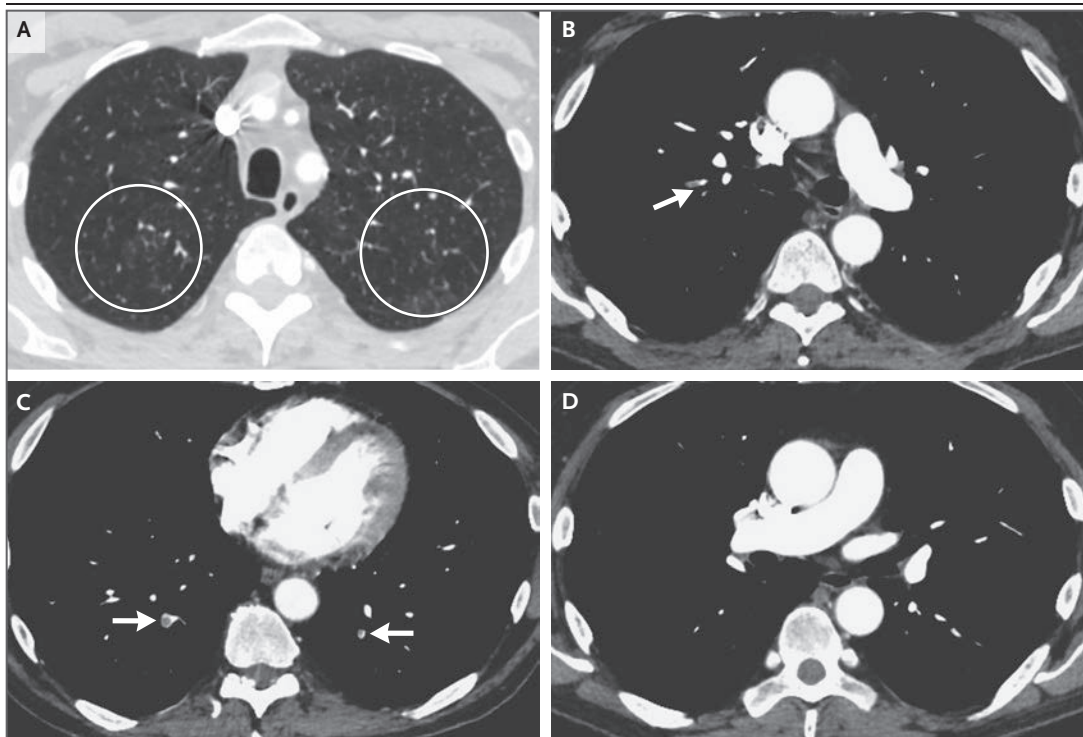


Figure 1. CT Pulmonary Angiogram Obtained on Presentation.

Mild tree-in-bud and ground-glass opacities are present in the upper lobes (Panel A, circled), findings that are probably indicative of inflammatory changes. Small filling defects are present in subsegmental pulmonary arteries of the right upper lobe (Panel B, arrow) and both lower lobes (Panel C, arrows), findings that are consistent with pulmonary emboli. The heart (Panel C) and pulmonary artery (Panel D) are normal, with no radiologic evidence of strain on the right side of the heart.

rin was initiated, and the patient was admitted to the hospital. On the second hospital day, the oxygen saturation was 91% while he was at rest and receiving supplemental oxygen through a nasal cannula at a rate of 5 liters per minute.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Eric P. Schmidt: This 67-year-old man with mantle-cell lymphoma and a history of pulmonary embolism was admitted to this hospital for hypoxemia that did not abate with the administration of supplemental oxygen. I will develop a differential diagnosis to identify the cause of hypoxemia, which can be broadly defined as a state of decreased oxygen content in the blood. If hypoxemia persists, mitochondrial delivery of oxygen is compromised, which leads to tissue hypoxia and organ dysfunction. Rapid assess-

ment and treatment of hypoxemia are therefore prioritized in the care of an acutely ill patient.

Lung disease is a common cause of hypoxemia. Indeed, this patient's complex medical history could suggest numerous possible pulmonary diseases. He had a history of two different cancers, and both diseases are capable of extension into the pulmonary circulation or lung parenchyma, although this would be uncommon. He was immunocompromised owing to the combined effects of cancer, ongoing chemotherapy, and remote splenectomy, so he was at risk for both community-acquired and opportunistic pulmonary infections. He had had multiple venous thromboembolic events in the context of cancer, so he was at risk for acute or chronic pulmonary embolism. Finally, it is important to consider whether his chemotherapeutic regimen could cause pneumonitis.

To narrow this broad differential diagnosis, it

Table 2. Five Physiological Mechanisms of Hypoxemia.*

Mechanism	Description	A-a Gradient†	Corrected with Supplemental Oxygen	Notable Feature
Low ambient atmospheric pressure	Decrease in atmospheric pressure leads to decrease in PAO_2 ‡	Normal	Yes	Diagnosis is usually obvious from patient's location at presentation
Hypoventilation	Increase in $Paco_2$ leads to decrease in PAO_2 ‡	Normal	Yes	Blood gas analysis shows arterial hypercarbia
Loss of diffusion capacity	Decrease in effective area for gas exchange leads to decrease in diffusion of oxygen into pulmonary circulation	Elevated	Yes	Hypoxemia worsens with exertion
Ventilation-perfusion mismatch	Increase in proportion of underventilated alveoli leads to increase in mismatch between ventilation and perfusion	Elevated	Yes	Process is often associated with airway disease
Shunting	Deoxygenated blood moves from right ventricle to left ventricle without encountering aerated alveoli	Elevated	No, if shunt fraction is sufficiently high	Shunt may be cardiac or pulmonary (alveolar)

* A-a denotes alveolar-arterial, FIO_2 fraction of inspired oxygen, $Paco_2$ partial pressure of arterial carbon dioxide, PAO_2 partial pressure of alveolar oxygen, and PaO_2 partial pressure of arterial oxygen.

† The A-a gradient is the difference between the PAO_2 (estimated with the alveolar gas equation) and the PaO_2 (measured with arterial blood gas analysis).

‡ This mechanism of hypoxemia is reflected in the alveolar gas equation: $PAO_2 = [\text{atmospheric pressure} - \text{water vapor pressure}] \times FIO_2 - Paco_2 / \text{respiratory quotient}$.

may be useful to turn to physiological principles. Rather than immediately attempting to determine the pulmonary diagnosis, I will first determine the specific physiological process that is likely to explain his hypoxemia. Generally, I will consider five major mechanisms of hypoxemia (Table 2).

REVIEW OF MECHANISMS OF HYPOXEMIA

Low Ambient Atmospheric Pressure

Any decrease in the atmospheric pressure leads to a decrease in the partial pressure of alveolar oxygen (PAO_2). There is nothing wrong with the lungs when this occurs. As such, the alveolar-arterial (A-a) gradient — the difference between the PAO_2 and the partial pressure of arterial oxygen (PaO_2) — is normal. This form of hypoxemia can be corrected by increasing the fraction of inspired oxygen (FIO_2) with the administration of supplemental oxygen. Because this patient lives at sea level, I can easily rule out a

decrease in atmospheric pressure as the cause of his hypoxemia.

Of note, this mechanism of hypoxemia is reflected in the alveolar gas equation: $PAO_2 = [\text{atmospheric pressure} - \text{water vapor pressure}] \times FIO_2 - Paco_2 / \text{respiratory quotient}$. ($Paco_2$ denotes partial pressure of arterial carbon dioxide.)

Hypoventilation

The alveolar gas equation is also useful in understanding the mechanism of hypoxemia in patients with hypoventilation. Carbon dioxide is freely diffused into alveoli. Thus, an increase in the $Paco_2$ leads to a decrease in the PAO_2 . There is nothing wrong with the lungs when this occurs, and the A-a gradient is normal. This form of hypoxemia can also be corrected by increasing the FIO_2 . Although results from arterial blood gas analysis are not provided for this patient, and arterial hypercarbia cannot be formally ruled out, there is no evidence in his history that

would suggest hypoventilation. Furthermore, in patients with hypoventilation, the oxygen saturation on pulse oximetry increases when supplemental oxygen is administered. Thus, hypoventilation cannot explain this patient's hypoxemia.

Loss of Diffusion Capacity

When hypoxemia is due to loss of diffusion capacity, the PAO_2 is adequate, but the ability of oxygen to diffuse into the arterial blood is limited or impaired. Loss of diffusion capacity occurs in patients who have conditions that are characterized by a decrease in the effective area for gas exchange, such as those who have undergone pneumonectomy (which decreases the absolute alveolar area for gas exchange) or those with pulmonary fibrosis (which decreases the effectiveness of the alveolar area for gas exchange). The process is defined by an elevated A-a gradient. This form of hypoxemia can be corrected with the administration of supplemental oxygen; a supranormal PAO_2 can overcome an elevated A-a gradient to adequately oxygenate the blood.

Loss of diffusion capacity is most apparent during exercise. When the speed at which erythrocytes move through alveoli is increased, the time available for gas diffusion is decreased, which amplifies the deleterious effects of impaired oxygen diffusion. In this patient, the oxygen saturation on pulse oximetry did not respond to an increase in the FI_{O_2} , and the hypoxemia worsened only modestly with exercise. Therefore, loss of diffusion capacity does not appear to be the primary mechanism of his hypoxemia.

Ventilation–Perfusion Mismatch

Ventilation–perfusion mismatch is a complex physiological mechanism. Even healthy people have heterogeneity in alveolar ventilation: many alveoli are well ventilated, but others are under-ventilated. This baseline inefficiency in ventilation may partly contribute to the presence of an A-a gradient in the context of normal airflow. In disease states that are characterized by impaired or disorganized airflow, the proportion of under-ventilated alveoli increases, which leads to an increased mismatch between ventilation and perfusion and ultimately to an elevated A-a gradient. Because under-ventilated alveoli are still

ventilated (and thus can be accessed by inhaled oxygen), this form of hypoxemia can be corrected with the administration of supplemental oxygen. This patient did not have apparent airflow abnormalities on examination, nor did he have a response to supplemental oxygen, so there was no clear evidence of ventilation–perfusion mismatch.

Shunting

Shunting is a concept that can be understood through consideration of congenital cyanotic heart disease. In patients with tetralogy of Fallot, deoxygenated blood in the right ventricle flows through a ventricular septal defect to the left ventricle, effectively bypassing aerated lung. The red cells never encounter oxygenated alveoli; the effect is similar to an elevated A-a gradient. Moreover, the red cells are unaffected by any amount of supplemental oxygen that is delivered to the alveoli because they will never encounter the oxygen. It is notable that small anatomical shunts within the bronchial circulation and Thebesian veins are present in healthy people and contribute to the presence of a normal A-a gradient.

Shunting also occurs when the pulmonary circulation delivers deoxygenated blood to a completely consolidated (and thus nonaerated) area of the lung. The blood flows from the right side of the heart to the left side without encountering aerated lung. In this patient, the oxygen saturation on pulse oximetry did not increase with the administration of supplemental oxygen, so shunting is the most likely mechanism to explain his hypoxemia.

Review of the mechanisms of hypoxemia suggests that this patient's presentation is most likely to be associated with shunting. My next task, therefore, is to find the source of the shunt. Could the patient have a substantial alveolar consolidation from an opportunistic infection or a drug-induced pneumonitis? Could he have acute right heart failure from pulmonary emboli that led to right-to-left shunting through a patent foramen ovale? The next step is to review the chest imaging in order to answer a specific question: where is the shunt?

REVIEW OF CHEST IMAGING

Review of this patient's chest imaging provides an opportunity to better integrate the history

and clinical findings with the imaging findings in order to establish a diagnosis. Although this patient had some scattered ground-glass opacities on chest imaging, this feature would not cause an intrapulmonary shunt sufficient to explain the lack of response to supplemental oxygen.¹ Furthermore, although he had pulmonary emboli, the normal size of the pulmonary artery and right ventricle suggests that a right-to-left pressure gradient of sufficient magnitude to open a patent foramen ovale is unlikely.

Thus, I am left with a discrepancy. The observed lack of response to supplemental oxygen on pulse oximetry suggests the possibility of a substantial physiological shunt, but no such shunt is apparent on imaging. This discrepancy prompts me to reassess whether the patient actually has a physiological shunt. Pulse oximetry is not infallible; it can be confounded by factors such as skin color² and hemoglobin abnormalities.³

METHEMOGLOBINEMIA

This patient was receiving dapsone, one of the most common acquired causes of methemoglobinemia.³ On pulse oximetry, methemoglobin has a red (660 nm) to infrared (940 nm) extinction coefficient, which is equivalent to that of 85% oxygenated hemoglobin.⁴ As the methemoglobin level approaches 30%, the oxygen saturation on pulse oximetry approaches 85% and is unaffected by supplemental oxygen, a process that mimics a physiological shunt.⁴ Arterial blood gas analysis by co-oximetry is not confounded by the presence of methemoglobin and thus can be used to quantify the methemoglobin level.

It is important to acknowledge that, on pulse oximetry, this patient had an oxygen saturation of 91 to 92% while he was at rest and receiving supplemental oxygen through a nasal cannula at a rate of 5 liters per minute. If this oxygen saturation is purely an artifact of methemoglobinemia, I would expect a methemoglobin level of less than 30% and no symptoms. However, this patient had dyspnea as well as anemia, which could be due to the effects of chemotherapy or could be dapsone-induced hemolytic anemia, a condition that is supported by the patient's elevated lactate dehydrogenase level and mean corpuscular volume. Methemoglobin levels as low as 8% have been associated with symptoms in patients with concurrent anemia.³ Further-

more, this patient had small pulmonary emboli, which are unlikely to explain his lack of response to supplemental oxygen but are likely to at least contribute to his dyspnea.

I suspect that the diagnosis is methemoglobinemia due to dapsone in a patient with concurrent pulmonary emboli. My diagnostic test would be arterial blood gas analysis by co-oximetry, which allows for methemoglobin quantification.

DR. ERIC P. SCHMIDT'S DIAGNOSIS

Dapsone-induced methemoglobinemia.

DIAGNOSTIC TESTING

Dr. Li Liu: The diagnostic test in this case was measurement of the blood methemoglobin level, which was 9.7% (reference range, 0 to 1.5). Methemoglobin is measured by the co-oximetry module of blood gas analyzers with characteristic absorbance spectra of the hemoglobin derivatives and is expressed as a percentage of the total hemoglobin level.

Normal hemoglobin contains iron in the ferrous state (Fe^{2+}) in the heme molecule. Oxidation of iron to the ferric state (Fe^{3+}) gives rise to an aberrant form of hemoglobin called methemoglobin. The ferric iron is unable to bind oxygen. In addition, the presence of ferric iron causes allosteric changes in the hemoglobin molecule to increase the affinity of the remaining ferrous iron for oxygen, thus leading to a left shift of the oxygen-hemoglobin dissociation curve. The dual effects of reduced oxygen release and decreased oxygen-binding capacity result in tissue hypoxia.^{5,6}

Red cells are under continuous exposure to oxidative stresses that convert hemoglobin to methemoglobin. However, a normal methemoglobin level is typically maintained through enzymatic reduction pathways, most importantly the cytochrome b_5 reductase pathway. Elevation in methemoglobin levels occurs with congenital enzyme deficiencies, hemoglobin variants (hemoglobin M disease), or exposure to exogenous oxidizing agents.^{5,6} Dapsone is a common medication that causes acquired methemoglobinemia.³ It induces oxidative stress through its highly reactive *N*-hydroxylamine metabolite, which can in turn oxidize hemoglobin to methemoglobin.^{3,6,7}

The most likely cause of methemoglobinemia in this patient is dapsone.

PATHOLOGICAL DIAGNOSIS

Dapsone-induced methemoglobinemia.

FOLLOW-UP

Dr. Hofacker: A transthoracic echocardiogram was obtained before the blood methemoglobin level was available. During echocardiography, the oxygen saturation on pulse oximetry was 94% while the patient was at rest and receiving supplemental oxygen through a nasal cannula at a rate of 6 liters per minute. After intravenous agitated saline was administered while he was at rest, there was evidence of a patent foramen ovale with trace right-to-left shunting. The right ventricular size and function were normal, with no evidence of acute right ventricular strain. The left ventricular systolic function was also normal.

The patient's anemia remained stable, and no packed red cells were transfused. Examination of a peripheral-blood smear revealed no evidence of hemolysis.

After the blood methemoglobin level became available, intravenous methylene blue was administered. Twelve hours later, the patient had been weaned off supplemental oxygen; the oxygen saturation on pulse oximetry had normalized, and repeat laboratory testing revealed that the methemoglobin level had decreased to 1.5%. The patient received cycle 4 of R-DHAP chemotherapy, which had been delayed during the evaluation for hypoxemia. The administration of dapsone was stopped, and atovaquone was restarted for prophylaxis against *P. jirovecii* pneumonia. The patient was instructed to avoid dapsone in the future.

The patient completed four total cycles of R-DHAP. Positron-emission tomography-CT and colonoscopy that were performed after treatment showed evidence of complete remission. Eighteen months later, the patient has had no evidence of disease recurrence.

Dr. Jacob A. Fox (Medicine): If the patient had had elevated pulmonary-artery pressures, would it have been safe to treat with methylene blue?

Dr. Schmidt: Methylene blue is a treatment for methemoglobinemia, but it is sometimes also used as salvage therapy for refractory vasoplegic

shock. By inhibiting soluble guanylyl cyclase (sGC; a downstream effector of nitric oxide), methylene blue prevents the production of cyclic guanosine monophosphate (cGMP) in vascular smooth muscle, thereby attenuating vasodilation. Although this effect may be beneficial in the treatment of systemic vasoplegia, blocking sGC could have untoward effects in patients with pulmonary hypertension. Indeed, pulmonary nitric oxide-sGC-cGMP signaling promotes pulmonary-artery vasorelaxation and thus may be beneficial in patients with pulmonary hypertension. Drugs that increase cGMP production (e.g., riociguat, an sGC activator) or prevent cGMP degradation (e.g., sildenafil, a phosphodiesterase-5 inhibitor) are commonly used in the management of pulmonary arterial hypertension and chronic thromboembolic pulmonary hypertension. Thus, blocking sGC with methylene blue might have unwelcome effects on pulmonary vasoconstriction, thereby worsening pulmonary hypertension.

Despite these theoretical concerns, animal studies suggest that methylene blue may have a minimal effect on pulmonary nitric oxide signaling.⁸ However, in practice, if this patient had had clinically significant pulmonary hypertension, I would have avoided the use of methylene blue. My decision would be driven less by the perceived risk of methylene blue and more by the lack of a convincing indication for treatment. Given this patient's relatively low methemoglobin level on repeat testing and the absence of organ dysfunction, it would be reasonable to simply stop dapsone therapy and provide supportive care.

Dr. Meridale V. Baggett (Medicine): Does the finding of trace right-to-left shunting through a patent foramen ovale on a transthoracic echocardiogram suggest that shunting contributed to this patient's hypoxemia?

Dr. Schmidt: If bubbles within agitated saline can pass from the right atrium to the left atrium without being intercepted by the pulmonary circulation, then shunting of deoxygenated hemoglobin is almost certainly occurring and would be contributing to hypoxemia. However, in this patient, only "trace" bubble transit was observed. Would it be sufficient to explain his lack of response to supplemental oxygen? Classic physiological "iso-shunt" diagrams show that a substantial shunt fraction (>25%) is necessary to

induce a lack of response to supplemental oxygen.⁹ Thus, I doubt that trace bubbles in the left side of the heart would indicate the presence of a shunt substantial enough to fully explain this patient's presentation. Furthermore, the use of methylene blue promptly resolved the observed hypoxemia, which is consistent with methemoglobinemia being the primary cause.

FINAL DIAGNOSIS

Dapsone-induced methemoglobinemia.

This case was presented at the Medicine Case Conference. Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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